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1.0 Valuation of Selvaag Bolig ASA

We have in this case performed a business valuation of Selvaag Bolig ASA. Our valuation is based purely on public information and is meant to be a comprehendible read. We have concluded that Selvaag Bolig is undervalued based on our findings.

1.1 Brief presentation of Selvaag Bolig

Selvaag Bolig ASA, hereby Selvaag Bolig, is a Norwegian development company which controls the entire value chain from acquisition of land to sale of homes. Selvaag Bolig offers a broad variety of property types and has several thousand homes under development at any given time. Selvaag Bolig has their focus on expanding in growth areas in the main cities in Norway which includes Oslo, Bergen, Stavanger, Trondheim and Stockholm in Sweden.

Selvaag Bolig is an industry leader in terms of profit margin and aims to pay a minimum 60 percent of its annual net profit as dividend. Selvaag Bolig was listed on the Oslo Stock Exchange on 14. June 2012. The biggest shareholder is Selvaag AS, the parent company of Selvaag Bolig, which owns 53,52% of the outstanding shares. The company is headquartered at Ullern in Oslo

2.0 Strategic Analysis

2.1 Macroeconomic analysis

For this part of the valuation, a macroeconomic analysis will be conducted.

2.1.1 Pest-analysis

The Pest-model analyses how *political/legal*, *economic*, *socio cultural* and *technological* environment impacts the firm's cash flow and risk. Within these environments we find different macroeconomic factors that affect industries and firms differently. We have therefore chosen to exclude macro factors that don't influence Selvaag Bolig.

Political/legal factors

Relevant political/legal factors for Selvaag Bolig consist of environmental protection law, taxation policy, employment law and government stability. Norway recently went through a change of government, and since Selvaag Bolig mainly operates in Norway, we may assume that these factors can become stricter in the future. This will affect Selvaag Bolig ASAs cash flow and risk a bit negative, but not worrying. Norway is altogether a stable country with a stable government. Selvaag Bolig ASA is therefore located in a good political/legal environment.

Economic factors

When it comes to the economic environmental factors, business cycles and GNP trends, interest rates, and inflation are what should be considered. Through Norges Bank, Norway adjusts its key interest rate to aim for an inflation target of 2% each year to continuously stabilise business cycles. However, during the corona pandemic, the key interest rate was set to 0 percent. Norges Bank plans to gradually increase the key interest rate back to its state before covid-19 to counter a possible coming economic upturn. This will have a negative effect on Selvaag Boligs cash flow and risk since a higher key interest rate leads to higher interest rate in the bank, which again leads to low demand from private households, meaning fewer people would want to invest in housing.

Sociocultural factors

Following the economic environmental factors, the sociocultural factors cover the income distribution, social mobility, and lifestyle changes. Selvaag Bolig ASAs cash flow and risk vary with these factors. Income distribution and social mobility is made possible through Norway's political and welfare system. This affects Selvaag Boligs cash flow and risk in a positive way, because a good income distribution leads to more people with wealth, giving Selvaag Bolig ASA more potential clients, and the possibility for more expensive projects. However, change in lifestyles, such as more people taking to de-urbanization, may affect Selvaag Boligs cash flow and risk negatively, unless they are able to adapt to the changes, then it will have a positive effect.

Technological factors

Finally, the technological environmental factors. Factors of relevance in this environment are government spending on research, new development, and speed of technology transfer(s) come into the picture. All these factors will have a positive effect on Selvaag Bolig ASAs cash flow and risk. If

done properly, projects might be completed faster due to better and more advanced technology and new development, which is positive for Selvaag Bolig.

2.2 External analysis

2.2.1 Porter's Five Forces - industry analysis

The attractiveness of an industry is defined by its capability to earn acceptable return. More competition reduces the chances of earning returns that exceed the cost of capital. By using Porter's Five Forces analysis, we can gain an understanding of the industry competition.

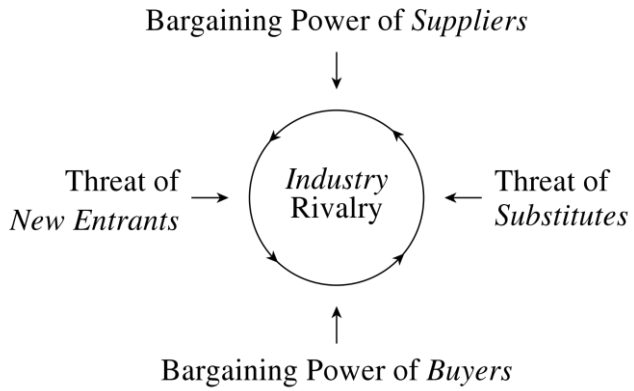


Figure 1: Porter's five forces analysis

Competitive Rivalry

The first of the five forces is competitive rivalry, and it considers the rivalry among existing competitors, to give an understanding of the level of competition. Tough competition tends to affect returns negatively (Plenborg & Kinserdal, 2021, p. 274). A company operating in an industry with a large number of competitors that offers equivalent products and services, will have less power (The Investopedia Team, 2020). According to Plenborg and Kinserdal (2021, p.274), competition tends to get intense if the industry is characterised by slow industry growth, numerous competitors, customers that perceive the product as commodities, overcapacity, and high exit barriers.
Selvaag Bolig operates within housing development, delivering turnkey properties[1] to clients. There are several competitors, with equivalent products and services to offer, within this industry, as well as high exit barriers, making the competition in the industry tough.

[1] Move-in ready homes.

Potential entrants

The second of the five forces is the potential of new entrants into the industry. New entrants seek to gain market shares, and often bring new capital to the industry (Plenborg & Kinserdal, 2021, p. 273). The less time and money it cost for a competitor to enter a company's industry and be an effective competitor, the more an established company's position could be significantly weakened (The Investopedia Team, 2020). These forces ultimately affect a company's return negatively. Typical entry barriers for new entrants, as stated in Plenborg and Kinserdal (2021, p.273), are economies of scale, product differentiation, capital requirements, switching costs, access to distribution channels and government policy. These barriers help protect a company's return.

As for the housing development industry, new entrants would require a lot of capital for housing projects, differentiated products to gain customers and be able to gain access to distribution channels. Based on this, Selvaag Bolig hits medium to low threat from new entrants since they already are developed within this industry.

Buyers

The third force is the bargaining power of buyers. This force reflects the relative strength between buyers and the industry. The potential return in the industry is typically limited if the buyers possess a high bargaining power (Plenborg & Kinserdal, 2021, p. 275). Buyers gain power through how many buyers there are, how significant each buyer is, how much it would cost the company to acquire new customers or for the company to find markets for its products (The Investopedia Team, 2020). Buyers are usually powerful if a large percentage of sales in an industry is purchased by a small percentage of buyers, the purchased products are commodities, low switching cost, experience low returns (Plenborg & Kinserdal, 2021, p. 275).

When it comes to buyers for Selvaag Bolig, we can conclude that they have low bargaining powers. In this industry there are a lot of buyers, costs for finding new buyers are insufficient, and switching costs are high.

Substitutes

Fourth of the five forces is the threat of substitutes. Products and services that can be used rather than a company’s, pose a threat. Companies that offer products and services that have no close substitutes will have leeway to increase prices and lock in favourable terms (The Investopedia Team, 2020). Substitutes limits the potential returns of an industry since high returns attract substitutes. Substituting products deserve more attention, according to Plenborg and Kinserdal (2021, p.274-

275), if they have potential to improve price-performance relation to already existing products in the industry and are produced by industries achieving high returns.
The goods and services offered by Selvaag Bolig cannot be that easily substituted, since most people prefer to have a home rather than living in, for example, a motorhome or even tents. There is close to no threat from substitutes for Selvaag Bolig.

Suppliers
Finally, the fourth of the five forces is the bargaining powers of suppliers. The more bargaining power suppliers have over participants in the industry, the more profitability can be squeezed from the industry by raising prices or by lowering the quality of the products or services offered. Suppliers tend to have more power, according to Plenborg and Kinserdal (2021, p.275), if they are more concentrated than the industry, the industry is not important for the supplier's outlet, the industry rely on the products and services offered by the suppliers, high switching costs for the industry. (Plenborg & Kinserdal, 2021, p. 275). Selvaag Bolig ASA rely heavily on their suppliers for their projects. The industry, in which Selvaag Bolig operates in, is also important for the supplier's outlet because there are not many other industries that need the same products and services that the suppliers offer. Suppliers have little bargaining power.

2.3 Internal analysis

2.3.1 SWOT-analysis

The SWOT-analysis is a fourfold analysis consisting of a company's internal strengths and weaknesses, and external opportunities and threats.

Strengths	**Weaknesses**
- Selvaag+	- Large inventory under development
Opportunities	**Threats**
- National expansion - International expansion	- De-urbanization - Raised key interest rate

Figure 2: SWOT-analysis

Strengths

Selvaag pluss is a strength for Selvaag Bolig because it is something only they offer. It's a household concept where Selvaag Bolig wants to bring hotellike service into their client's residence through extra service, experiences, and safety.

Weaknesses

Since Selvaag Bolig operates within housing development, a large inventory is almost inevitable. Large sums are tied up to the inventory because of undeveloped land and work in progress projects. This is a weakness in scenarios where the market is experiencing a downturn, forcing Selvaag Bolig to amortise their inventory.

Opportunities

Selvaag Bolig ASA is already well established in Norway with a lot of houses and major projects in the biggest cities, as well as counties. An opportunity for Selvaag Bolig is to expand even further in Norway than what they have today, to even more cities and counties. Not only can they expand in Norway, but on an international basis. They have already some projects in Sweden and Spain, so a further international expansion is not impossible.

Threats

Raised key interest rate poses a threat for Selvaag Bolig. Raised key interest rate will have a negative macroeconomic effect on housing prices because of more expensive bank loans.

3.0 Analysis of accounting information

3.1 Assessment of accounting quality

"*Good accounting policy is characterized by financial statements that provide an objective (neutral) picture of a firm's financial position and is free of manipulation*" (Petersen et al., 2019, p. 334).

Some of the main users/stakeholders and their uses of a company's accounting are: 1. equity-oriented - the need to estimate the value of the firm, 2. Debt-oriented - lenders, and 3. compensation-oriented. In order for the users to assess the company's accounting, a list must be examined. The list consists of motives for accounting manipulation, the quality of applied accounting

policies, permanent versus transitory accounting items, the level of information in the financial statements, and identification of red flags.

Assessment of the following statements	TRUE 1	2	3	4	FALSE 5
Analysis of accounting quality has generally shown:					
Few motives for accounting manupilation	X				
A high degree of quality in accounting policies	X				
A high degree of recurring accounting items		X			
A high lever of information in the annual report	X				
Few 'red flags' in the financial statement	X				
Total appraisal	**1,2**				

Figure 3: Assessment of accounting quality.

As shown on the assessment above, we see that Selvaag Bolig ASA has an overall good accounting quality. The only thing that was striking, was that the post *Goodwill* from the *Balance Sheet*, as it was constant through all the analysis years.

3.2 Reformulating the Financial Statements

In order to calculate the different key figures to further analyse the firm at a satisfactory level, both the income statement and balance sheet have to be reformulated. By reformulating the given financial statements, we obtain the necessary information to analyse the company's profitability, liquidity, and associated growth rates.

3.2.1 The analytical income statement

When reformulating the income statement, we firstly divide the statement into two categories: Operating and financial. We then arrange it accordingly to calculate NOPAT, (net operating profits after taxes). To do this we need to locate all the operating income and expenses from the income statement. We chose to extract the following post from the statement: ***Other gains (losses), net.***[2] Our reason for doing so is that we recognise this post as a non-recurring post, and we therefore find it irrelevant for our analysis due to it being an extraordinary income and only occurring in the years 2016 and 2020.

[2] 2016 (31,68 MNOK) and 2020 (1.045MNOK).

	2016	2017	2018	2019	2020
Revenue	3 000 347	3 228 832	3 342 075	3 368 838	2 697 950
Gross profit	620 601	796 014	920 442	1 089 353	730 364
EBITDA	382 285	547 737	754 187	877 455	608 481
EBIT	362 224	543 653	750 480	864 707	598 678
NOPAT	276 084	415 282	581 412	672 109	469 095
Net income	250 119	388 831	567 681	663 727	480 945

In 2020, the company's revenue took a considerable hit and discontinued the growth in revenue from previous years. This can be explained by the company's method of accounting, where they have stated the following in their annual report:

"The group utilises the percentage of completion method for revenues and cost of goods sold, where the degree of completion is estimated on the basis of expenses incurred relative to total estimated cost multiplied by the sales rate. "

Thanks to this method, the decline in revenue was severely affected by the pandemic. This resulted in fewer units completed which was caused by a shortage of labour mainly due to the pandemic.

3.2.1 The analytical balance sheet

When reformulating the balance sheet, the posts are also divided into two different categories: operating and financial. To calculate NONCA[3], we (re)arrange the values and move all the values that are related to operations onto the left side. We follow the same path to find NOWC[4]. By adding the two of them together, we get NOA[5], which is the value we have been looking for.

To calculate the right side (of the sheet), we add all the interesting bearing debt together and move the cash post to the right (hand) side, where we then get NIBD[6]. We then add NIBD with the company's Total Equity and the analytical sheet is (finally) balanced. The reasoning behind this is to

[3] NONCA: Net Operating Non-Current Assets
[4] NOWC: Net Operating Working Capital
[5] NOA: Net Operating Assets
[6] NIBD: Net Interest-Bearing Debt

get a more accurate overview of Selvaag Bolig ASA´s operational and financial statements.

	2016	2017	2018	2019	2020
NONCA	778 639	895 807	1 095 976	976 455	890 247
NOWC	3 608 275	3 694 848	3 679 709	3 481 291	3 131 693
NOA	4 386 914	4 590 655	4 775 685	4 457 746	4 021 940
Equity	2 699 172	2 858 142	3 116 136	3 382 084	2 437 815
NIBD	1 687 742	1 732 513	1 659 549	1 075 662	1 584 125
Total	4 386 914	4 590 655	4 775 685	4 457 746	4 021 940

Net Operating Non-current Assets (NONCA)

The reason NONCA decreased from 2018 and onwards is mainly because of other non-current assets. From 2018 a decrease in seller credits is related to sales of shares[7]. There was also a small decrease in property, plant & equipment in 2019 and 2020. Mainly due to the big portion of depreciation in said post. In 2020 there weren't any receivables from sale of land, unlike the prior years.

Net Operating Working Capital (NOWC)

NOWC decreased over the years 2019 and 2020, mainly due to a decrease in Inventory Property caused by less developed land and fewer finished projects. There was also a small decrease in trade receivables.

Net Interest-bearing Debt (NIBD)

We can spot a noticeable decrease in NIBD in the year 2019 due mainly to a high increase in deposited cash. A decrease in non-current interest-bearing liabilities is seen in the years 2019 and 2020 and is also partly to blame.

[7] Sale of shares in the Kaldnes Brygge project, Tønsberg. Generated the following: 2017(118 MNOK), 2018 (86 MNOK), 2019 (64 MNOK), 2020 (54 MNOK).

3.3 Profitability Analysis

3.3.1 ROIC - Return on Invested Capital

"Return on invested capital (ROIC) measures the profitability of the operations. The ratio expresses the return on capital invested in a firm's net operating assets (NOA) as a percentage" (Plenborg and Kinserdal, 2021, 144).

$$ROIC_t = \frac{NOPAT_t}{\frac{NOA_{t-1}+NOA_t}{2}}$$

We chose to use average Net Operating Assets (NOA) to provide a more accurate output of ROIC. The ROIC then expresses the return on average invested capital*.

We do not, however, know if profitability comes from a good revenue and expense ratio or if it comes from improved capital utilisation. This is where the two components behind ROIC come into play. Firstly, we have the profit margin, which explains the ratio between operating income and net revenue. Profit margin is calculated by how much Net Operating Profits (NOPAT) make-up of the net revenue in percent. Second is the assets turnover rate which captures how the firm utilises its capital in form of how much revenue they generate out of the firm's total assets. In other words, how many Norwegian crowns[8] (NOK) the firm can generate from investing one Norwegian crown.

	2017	2018	2019	2020
ROIC before tax	12,11 %	16,03 %	18,73 %	14,12 %
ROIC after tax	9,25 %	12,41 %	14,56 %	11,06 %
Profit margin before tax	16,84 %	22,46 %	25,67 %	22,19 %
Profit margin after tax	12,86 %	17,40 %	19,95 %	17,39 %
Turnover rate	0,68	0,67	0,73	0,63

**The year 2016 is removed because of the use of average values.*

We can observe that the growth in ROIC comes from an increase in profit margin. The growth in profit margin is driven by a growth in revenues and low depreciation and amortisation costs. In 2020 we can observe that the ROIC decreases, and this comes from the decline in revenues in the same

[8] NOK: Norske Kroner. The national currency used in Norway. NOK/USD was 0,11 as of 31.12.2020.

year. To better understand the ROIC we will further compare it with the Weighted Average Cost of Capital (WACC) in the next section.

3.3.2 WACC - Weighted Average Cost of Capital

WACC is the required rate of return from shareholders and is the lowest expected return from a shareholder's perspective.

	2017	2018	2019	2020
ROIC after tax	9,25 %	12,41 %	14,56 %	11,06 %
WACC	4,61 %	4,84 %	5,59 %	5,48 %

If we get a ROIC value higher than the cost of capital, that will tell us that our company is experiencing growth and is overall a healthy company. Opposite will a ROIC value lower than the cost of capital suggests the opposite. We can observe that Seelvaag's ROIC is higher than their WACC every year which means that they have positive Economic Value Added (EVA) also known as super profit is shown below.

	2017	2018	2019	2020
ROIC after tax	9,25 %	12,41 %	14,56 %	11,06 %
WACC (Post Tax) Adjusted	4,61 %	4,84 %	5,59 %	5,48 %
NOA	4 590 655	4 775 685	4 457 746	4 021 940
EVA	213 187	361 623	399 861	224 595

This tells us that the required rate of return is higher than the actual return on capital. Which in turn means that Selvaag is creating value for their shareholders each year and throughout the calculated historical period.

3.3.3 ROE - Return on Equity

Return on equity (ROE) measures the profitability taking into account returns earned on operations plus the effect of financial leverage (Plenborg and Kinserdal, 2021, 170).

$$ROE = \frac{Net\ earnings\ after\ tax}{Book\ value\ of\ equity}\ x\ 100$$

Return on Equity is an important key figure for the shareholders and investors since it tells us how the company performs. To calculate ROE, we divide net earnings after tax on the book value of equity. Since shareholders take on bigger risk than the creditors ROE should be higher than ROIC and because of this, shareholders are expecting a bigger return. We can see from our calculated ROIC from earlier, that this is the case. To see if ROE is satisfying, we can compare it to the required return on equity (required ROE).

	2017	2018	2019	2020
ROE	14 %	19 %	20 %	17 %
Re	6,25 %	6,25 %	6,25 %	6,25 %

As seen above the ROE is higher than the required rate of return (Re) every and each year which means Selvaag generates a positive residual income every year.

3.4 Growth analysis

In order to track growth in selected key figures, growth analysis is essential, both for an organisation and for its stakeholders. The growth in these key figures will interest lenders, suppliers, employees, etc. The growth analysis is essential to get a picture of the company's ability to create value and to determine if it's a growth company.

	2017	2018	2019	2020
Revenue	7,62 %	3,51 %	0,80 %	-19,91 %
EBIT	50,09 %	38,04 %	15,22 %	-30,77 %
Net income	55,46 %	46,00 %	16,92 %	-27,54 %
NOA	4,64 %	4,03 %	-6,66 %	-9,78 %
Equity	5,89 %	9,03 %	8,53 %	-27,92 %
NIBD	2,65 %	-4,21 %	-35,18 %	47,27 %
EVA	385,11 %	91,28 %	22,19 %	-44,50 %

From looking at the table above we can observe that the values differentiate from year to year. Growth in revenue is often directly related to the market as we can observe from the year 2020 when the pandemic hit globally which also affects the company's Earnings Before Interest and Taxes (EBIT) and Net income. It led to a decrease in buying of apartments, houses etc.

As previously mentioned, a decrease in Net Operating Non-current Assets (NONCA) is the reason for the declining growth in Net Operating Assets (NOA). If we were to look at this from an owner's perspective the years prior to 2020 were solid because of the growth in equity and the abnormally declining interest-bearing debt. This is especially good for the shareholders because it means less

interest payments and higher net income and cash flows. To further analyse the growth, we will look at the sustainable growth rate below.

3.4.1. The sustainable growth rate

"Sustainable growth rate indicates at what pace a firm can grow its revenues while preserving its financial risk, i.e. maintain its financial leverage (debt to equity ratio) at the same level" (Plenborg and Kinserdal, 2021, 189).

$$g = \underbrace{\left[\text{ROIC}}_{\text{Operations}} + \underbrace{(\text{ROIC} - \text{NBC}) \times \frac{\text{NIBD}}{\text{E}}\right]}_{\text{Financial leverage}} \times \text{Minority interests share} \times \underbrace{(1 - \text{PO})}_{\text{Dividend policy}}$$

When measuring growth in a company the sustainable growth rate is the most used and is affected by three components: Operating profitability, financial leverage, and dividend policy. This makes it a great way to gain an overview of where the growth originates from.

	2017	2018	2019	2020
Net financial expenses after tax	64 172	34 439	33 526	14 211
NBC	3,70 %	2,08 %	3,12 %	0,90 %
Dividens paid (in NOK 1000)	242 179	308 850	388 567	1 422 714
Payout ratio	62,28 %	54,41 %	58,54 %	295,82 %
Retention ratio	37,72 %	45,59 %	41,46 %	195,82 %
Sustainable Growth - g	4,76 %	8,17 %	7,54 %	34,60 %

As we have seen throughout the report, the year 2020 differentiates a lot and has abnormal (growth) values. The sustainable growth rate increases every year but as for 2020 Selvaag paid out a substantially big dividend and this resulted in a relatively high sustainable growth rate, which will most likely not be recurring in the future. Selvaags ROIC is higher than NBC as per 2020 and if it stays that way this will positively affect the sustainable growth rate.

3.5 Liquidity analysis

Liquidity is crucial for a company as it is the ability to convert an asset into cash easily and without losing money against the market price. The easier you can turn an asset into cash, the more liquid the asset is. When analysing a company's liquidity, it's normal to separate between short-term and long-term liquidity risk. Short-term liquidity risk refers to the risk associated with obligations which

are to be paid within a year, and long-term liquidity risk refers to the risk associated with all future obligations, above a year or more.

3.5.1 Short-term liquidity risk

In our assessment of Selvaag's short-term liquidity risk, we have chosen to highlight the current ratio, quick ratio, the cash burn rate, and CFO to short-term debt. The current ratio is a liquidity ratio that measures a company's ability to pay short-term obligations, often within one year. Selvaags current ratio has been much higher than one between 2016-2020 which indicates that Selvaag has been in a good position to pay short-term obligations. The ratio has been reduced over the years but is still considered high compared to what is considered to be a good ratio.
Selvaag's quick ratio has been under one between 2016-2020. A normal quick ratio is often considered to be around one. The fact that Selvaag has had a quick ratio under one for the last years, indicates that Selvaag has room for improvement on their quick ratio.
Selvaag's Cash-burn rate is a measurement on how fast cash is being used and we have calculated it by taking cash plus the sum of trade receivables and other current receivables and dividing this on EBIT. Selvaag's positive cash-burn rate indicates that they are using reserves. This is not necessarily a bad thing, considering a positive cash-burn rate can help the firm grow.

Short term risk	2016	2017	2018	2019	2020
Quick Ratio	0,78	0,44	0,66	0,94	0,44
Current Ratio	3,63	3,53	3,68	2,51	2,15
Cash Burn Rate	3,26	1,21	1,24	1,52	1,68
CFO to short-term debt	N/A	0,14	0,28	0,43	0,39

3.5.2 Long-term liquidity risk

Financial leverage is the ratio of debt to total assets of the firm, which means what percentage of total assets is financed by debt. Selvaag's financial leverage was reduced from 2016 to 2019, before it increased in 2019. We believe that the increase is connected to Covid-19 that appeared in 2020. Selvaag's financial leverage is not on a critical level, but there's room for improvement.
The cash flow to debt ratio measures the extent to which a cash flow is sufficient to pay liabilities. A high ratio is desirable as it indicates a low long-term liquidity risk. Selvaag's ratio has improved from 2016 to 2020, but we still think the ratio is quite low and therefore indicate a financial risk. Selvaag has a room for improvement and a higher ratio would be preferable.

Long term risk	2016	2017	2018	2019	2020
Financial leverage	1,37	1,22	1,08	1,04	1,45
Solvency ratio	0,421	0,450	0,480	0,489	0,408
CFO To Debt Ratio	N/A	0,06	0,11	0,29	0,26

4.0 Forecasting

4.1. Forecasting Income statement

In our report, we have based our revenue growth on the information from the annual report. Our forecasted growth for 2021 is considerably higher than the following years and the reason for this is because the revenue of 2020 was weak due to the pandemic. We expect that the revenues for 2021 will climb back up to a higher level which results in a high revenue growth. The remaining financial drivers are divided into ratios as a percentage of revenue. The value driver of depreciation is calculated as a percentage of Selvaag's PP&E. The tax rate used is the average tax rate for the previous years, which have been exactly 22% each year. Below is a summary of some important numbers that have been forecasted. For further elaboration in the numbers, see appendix.

	2021	2022	2023	2024	2025	Terminal 2026
NOPAT	527 922	536 088	554 042	542 448	560 356	579 235
Growth NOPAT	12,54 %	1,55 %	3,35 %	-2,09 %	3,30 %	3,37 %
Net income	544 682	560 542	587 005	585 020	613 954	637 968
Growth Net Income	13,25 %	2,91 %	4,72 %	-0,34 %	4,95 %	3,91 %

4.2. Forecasted balance sheet

Goodwill is expected to be the same in the upcoming years since it has been the same amount the last five years. PP&E, deferred tax liabilities, trade payables, trade receivables, other receivables, other non-current bearing liabilities and other current non-interest-bearing liabilities is all calculated as a percentage of revenue. In most of the forecasting assumptions we have used the average of the last five years (2016-2020).

	Explicit					Terminal
	2021	*2022*	*2023*	*2024*	*2025*	*2026*
NONCA	999 307	1 016 068	1 044 354	1 072 657	1 102 091	1 117 397
NOWC	3 290 976	3 355 463	3 487 015	3 623 829	3 766 115	3 840 104
NOA	**4 290 283**	**4 371 530**	**4 531 369**	**4 696 486**	**4 868 207**	**4 957 502**
EQUITY	2 535 328	2 662 094	2 821 820	2 991 129	3 170 597	3 360 832
NIBD	1 684 537	1 573 401	1 630 930	1 690 359	1 752 165	1 784 304
TOTAL E + NIBD	**4 219 864**	**4 235 495**	**4 452 750**	**4 681 488**	**4 922 762**	**5 145 136**

IBD is calculated as a percentage of NOA. Net financial expenses have been calculated as a negative growth and have been declining the last year. We are assuming Selvaag is paying down its debt in the upcoming years and that the financial expenses will reduce for each year. In our forecasting we have calculated growth in total IBD instead of calculating each component that is included in IBD. Total Equity is also forecasted as an average growth from the last years. Non-controlling interest is forecasted as a constant and Equity attributable to shareholders of the company is calculated as Total equity minus non-controlling interest.

4.3. Forecasted Free cash flow

	2021	2022	2023	2024	2025
NOPAT	527 922	536 088	554 042	542 448	560 356
+Depreciation Expense	9 892	10 086	10 482	10 893	11 321
-/+ incr./decr. in NOWC	64 486	131 552	136 814	142 287	73 989
-/+ incr./decr. in NONCA (***incl. depreciation***)	(16 955)	(28 682)	(28 714)	(29 862)	(15 528)
FCFF	**585 346**	**649 044**	**672 624**	**665 766**	**630 137**
+/- incr./decr. in NIBD ex. cash	(169 828)	87 911	90 814	94 447	49 112
Financial income (4) (F)	23 444	28 133	33 760	40 512	48 614
Financial expenses (4) (F)	(9 707)	(8 089)	(6 741)	(5 617)	(4 681)
Tax-shield from NFE	3 022	4 410	5 944	7 677	9 665
FCFE	**432 278**	**761 409**	**796 402**	**802 784**	**732 847**
- (super) dividends	(417 916)	(400 817)	(417 696)	(405 552)	(423 718)
Cash surplus in period	**14 362**	**360 593**	**378 705**	**397 232**	**309 129**

By forecasting the financial statements, we could calculate the forecasted free cash flow statement. FCFF stands for free cash flow to the firm and FCFE stands for free cashflow to equity. FCFF is the operational cash flow and FCFE is the shareholders profit after financial expenses is paid to the creditors.

5.0 Valuation of Selvaag ASA

5.1. Estimated WACC

To calculate WACC we first must calculate the required rate of return on equity (Re) by using the Capital Asset Pricing Model (CAPM) shown below.

$$Re = Rf + \beta e * (Rm - Rf)$$

$Re = Required\ rate\ of\ return\ on\ Equity$
$Rf = Risk - free\ rate$
$\beta e = Equity\ beta$
$Rm = Expected\ return\ of\ the\ market$
$(Rm - Rf) = Equity\ market\ premium$

We retrieved the Equity Market Premium of **4,24%**, the Risk-Free Rate of **2,77%** and the Equity Beta of **0,82** and set these as constants over the historical period since we believe these values won't fluctuate noteworthy which gave us a Required rate of return on Equity of **6,25%** for all coming years.

Once we calculated the Required Return on Equity (Re) we found the Required Return on Interest Bearing Debt in the annual report(s) (Rd). We then calculated the Market Value of Equity by retrieving the weighted average of outstanding shares from the annual report(s) dividing it on the price per share[9] as shown below.

	2017	2018	2019	2020
Outstanding shares per 31.12	92 149 762	92 403 798	92 963 716	93 327 977
Price per share per 31.12	15,81	20,64	41,44	49,80
Market value of equity (in NOK 1 000)	1 456 888	1 907 214	3 852 416	4 647 733

We could then calculate the estimated Weighted Average Cost of Capital (WACC). As mentioned WACC is the required rate of return from shareholders and is the lowest expected return from a shareholder's perspective. We got the following estimates.

	2017	2018	2019	2020
WACC (Post Tax) Adjusted	4,61 %	4,84 %	5,59 %	5,48 %
WACC Fixed	5,48 %	5,48 %	5,48 %	5,48 %

[9] (Yahoo, 2022)

5.2. The present value method

It is through the valuation of a company that a clear picture of its present value is painted. Unlike a company's market value, our models provide a detailed analysis of the company's future cash flows, earnings, and growth prospects.

Fundamental valuation allows one to determine if the business is over- or undervalued, thus determining whether it is a good time to sell or buy the company's stock. The beta value, as mentioned earlier, was set to 0,82. Changing the beta value would radically affect the Weighted Average Cost of Capital (WACC) and therefore the valuation of the company. A buy or sell recommendation for Selvaag is therefore fraught with medium/high uncertainty.

5.2.1 The economic value added (EVA) method

We evaluated Selvaag using the EVA model and a DDM model with a Required Return on Equity as a constant. The EVA analysis is a method of valuing a company's financial performance by eliminating its cost of capital from its operating profit. This method attempts to estimate the "real" profit of a business. The method differs from the DDM valuation method which analyses future stock prices and is based on cash flow profit.

The EVA model calculates the value of the business as the current total capital plus the present value of future excess returns. By deducting NIBD from the discounted estimated Enterprise Value (EV), we end up with the market value of equity (MVE).

EVA (Fixed WACC)		Explicit forecasting period				Terminal
Period	0	1	2	3	4	5
	2020	2021	2022	2023	2024	2025
NOA	**4 021 940**	4 290 283	4 449 336	4 612 309	4 780 663	4 868 207
WACC (Post Tax)	5,48 %	5,48 %	5,48 %	5,48 %	5,48 %	5,48 %
NOPAT		**527 922**	**545 641**	**563 978**	**552 201**	**560 356**
Cost of Capital		220 392	235 096	243 812	252 742	261 968
EVA		**307 531**	**310 545**	**320 166**	**299 459**	**298 388**
Discount Factor		0,948	0,899	0,852	0,808	0,766
PV of EVAs in explicit at 31.12.0		**291 555**	**279 117**	**272 815**	**241 914**	
PV of EVAs in terminal at 30.12.5						8 575 029
PV of EVAs in terminal at 30.12.0						**228 526**
PV of all EVAs per 30.12.0	**1 313 927**					
EV per 30.12.0	5 335 867					
NIBD per 30.12.0	1 584 125					
MVE per 30.12.0	**3 751 742**					
Outstanding shares	93 327 977					
MV per share	**kr 40,20**					

5.2.2 Dividend Discount Model (DDM)

The DDM model uses dividend, where we use required return on equity (Re) as the discounting factor. The terminal period is the future value for all future cash flows in the years following our forecasted period. The DDM model we calculated is shown below.

$$P = \frac{D_1}{r - g}$$

P = stock price
D_1 = value of next year dividend
r = constant cost of equity capital
g = constant growth rate in perpetuity

		Explicit forecasting period				Terminal
Period	0	1	2	3	4	5
	2020	2021	2022	2023	2024	2025
Dividends	1 422 714	417 916	410 370	427 632	415 305	423 718
Re	6,25 %	6,25 %	6,25 %	6,25 %	6,25 %	6,25 %
Discount factor		0,941	0,886	0,834	0,785	
PV of terminal per 30.12.4					9 977 351	
PV of terminal per 30.12.0					**7 829 820**	
PV of explicit per 30.12.0		**393 345**	**363 533**	**356 551**	**325 914**	
MVE per 30.12.0	**9 269 164**					
Outstanding shares	93 327 977					
MV per share	**kr 99,32**					

We can observe the results from the EVA- and DDM-analysis in the table below. The market values we have estimated, suggests a price target of NOK 69,76 per share indicating that Selvaag Bolig ASA is underpriced by around NOK 1,8 billion since their market value is approximately NOK 4,67 billion as of 31.12.2020. The target price is somewhat unreliable since the values from EVA and DDM differentiate.

	NOK	Outstanding shares	Price per share
Market Value (EVA)	kr 3 751 742 319	93 327 977	40,20
Market Value (DDM)	kr 9 269 163 516	93 327 977	99,32
Market Value (Real)	kr 4 647 733 255	93 327 977	49,80
AVERAGE EVA +DDM	kr 6 510 452 918	93 327 977	69,76

6.0. Conclusion

We conclude the analysis with a buy recommendation, based on a price target of NOK 69,76. There are many variables that affect the company's current market value and it should be pointed out that there is a high degree of uncertainty associated with the calculation of the company's cost of capital (WACC). WACC influences the valuation to a large extent, in that it affects the company's present value. Although there are variables in the valuation that create some uncertainty about the business value of Selvaag Bolig ASA, we recommend buying their securities. Selvaag Bolig ASA is a leading real estate company and is already differentiating themselves from their competition. Which we believe will lead to further growth. Our conclusion is therefore a buy recommendation of Selvaag Bolig ASA's securities with a price target of NOK 69,76.

7.0 References

Damodaran, A. D. (2022, January 1). Country Default Spreads and Risk Premiums. Stern NYU. Retrieved May 13, 2022, from https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/ctryprem.html

Inflasjon. (n.d.). Norges Bank. https://www.norges-bank.no/tema/pengepolitikk/Inflasjon/

Porters five forces. (n.d.). [Graphical]. Wikipedia. https://en.wikipedia.org/wiki/Porter%27s_five_forces_analysis

Plenborg, T., & Kinserdal, F. (2021). Financial Statement Analysis (2nd ed.). Fagbokforlaget.

TRADING ECONOMICS. (n.d.). Norway Government Bond 10Y - 2022 Data - 1988–2021 Historical - 2023 Forecast - Quote. Retrieved May 15, 2022, from https://tradingeconomics.com/norway/government-bond-yield

The Investopedia Team. (2020, February 21). Porters 5 Forces. Investopedia. Retrieved June 2, 2022, from https://www.investopedia.com/terms/p/porter.asp

Yahoo. (2022). Selvaag Bolig ASA (SBO.OL). Yahoo Finance. Retrieved May 16, 2022, from https://finance.yahoo.com/quote/SBO.OL/?guccounter=1

Petersen, C. P., Plenborg, T. P., & Kinserdal, F. K. (Eds.). (2019). Accounting Quality, Chapter 13. In Financial Statement Analysis (1st ed., Vol. 1, pp. 333–372). Fagbokforlaget.